

Projeto

Tapejara

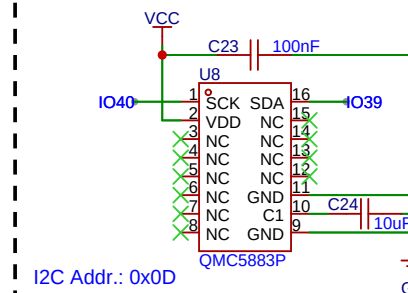
TPJ-01

Open Source Project

Rotors: coreless 8520 (2x CW + 2x CCW)
Propellers Diameter: 55 / 65 mm (2x CW + 2x CCW)
Battery: 1S (3.7V) | 500 mAh
PCB (X-Frame):
- Material: FR-4
- Color: Green
- Thickness: 1.2 mm

Remote Control: Mobile App

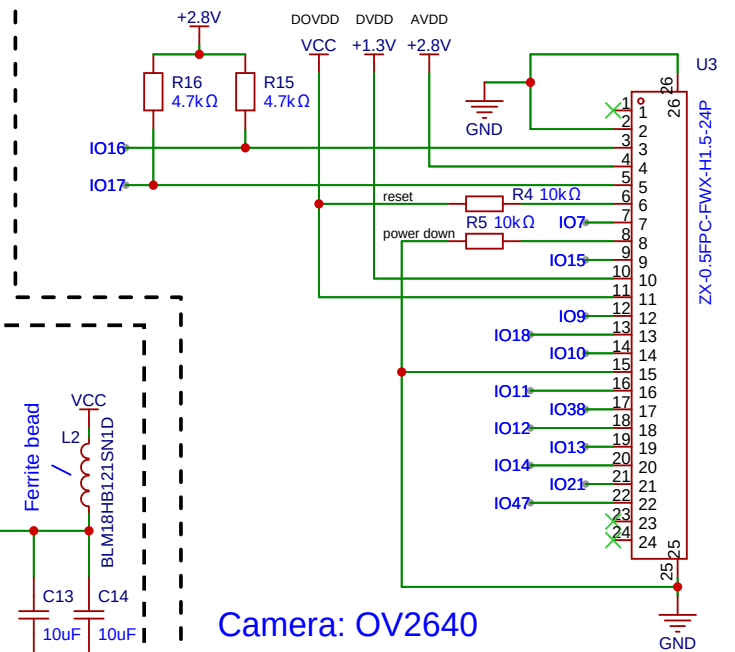
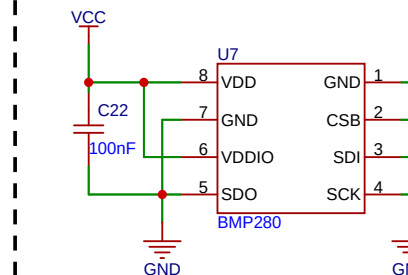
Magnetometer / Compass



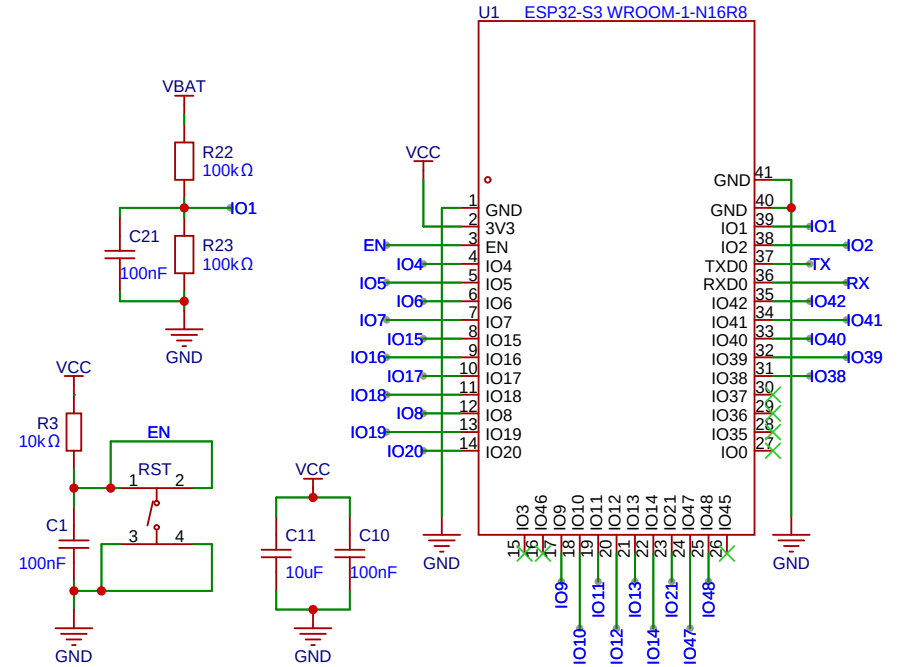
MAPEAMENTO CONECTOR CAMERA (24P) -> ESP32-S3-WROOM-1 (N16R8)				
PIN	NOME	FUNÇÃO	GPIO	OBSERVAÇÕES
01	STROBE	Não Conectado		
02	AGND	Terra Analógico		
03	SDA	SCCB Data (I2C)	16	Pull-up necessário (ex: 4k7 para 3.3V)
04	AVDD	Alimentação Analógica		2.8V nominal
05	SCL	SCCB Clock (I2C)	17	Pull-up necessário (ex: 4k7 para 3.3V)
06	RESET	Reset		Ativo em nível BAIXO (Low)
07	VSYNC	Vertical Sync	07	Sincronismo de quadro
08	PWDN	Power Down		Ativo em nível ALTO (High)
09	HREF	Horizontal Reference	15	Sincronismo de linha
10	DVDD	Alimentação Core		OV2640: 1.3V OV5640: 1.5V
11	DOVDD	Alimentação Digital IO		Conectar a 3.3V
12	D7 (Y9)	Dados Bit 7 (MSB)	09	
13	XCLK	External Clock	18	Clock de entrada (geralmente 24MHz)
14	D6 (Y8)	Dados Bit 6	10	
15	DGND	Terra Digital		
16	D5 (Y7)	Dados Bit 5	11	
17	PCLK	Pixel Clock	38	Clock de saída do sensor
18	D4 (Y6)	Dados Bit 4	12	
19	D0 (Y2)	Dados Bit 0 (LSB)	13	
20	D3 (Y5)	Dados Bit 3	14	
21	D1 (Y3)	Dados Bit 1	21	
22	D2 (Y4)	Dados Bit 2	47	
23	NC	Não Conectado		
24	NC	Não Conectado		

PINAGEM CABO FLAT 24 VIAS (OV2640 / OV5640) - PASSO 0.5MM

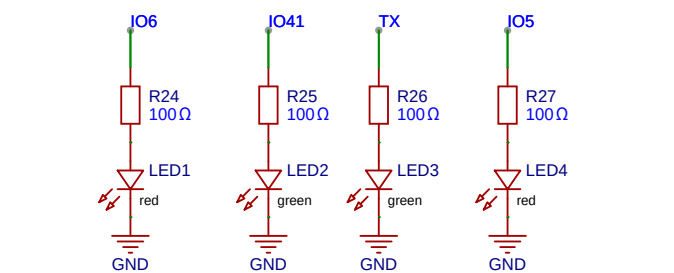
Barometer



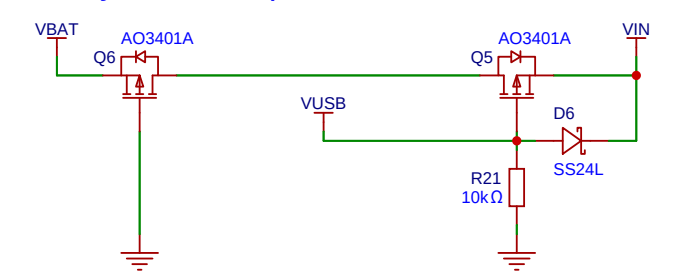
ESP32-S3 / Reset



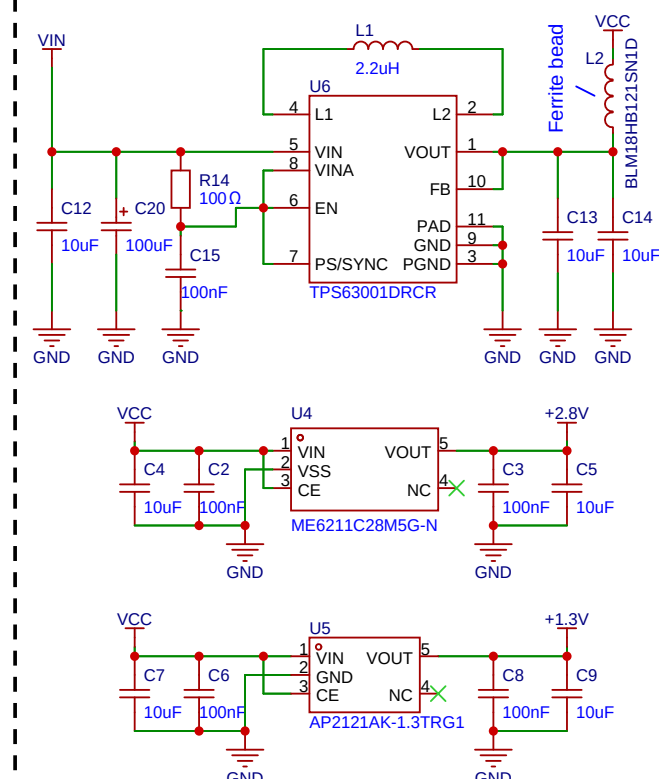
Sinalization



Voltage source selection + Battery inversion protection

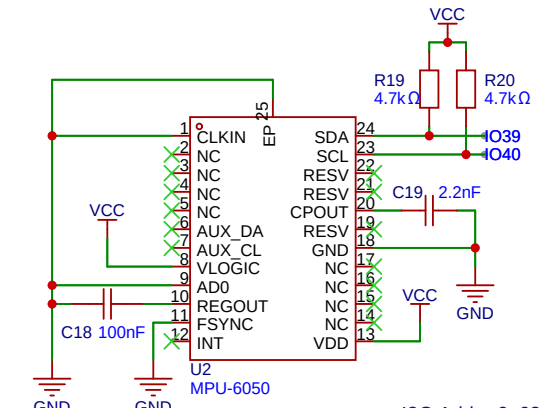


Voltage Regulators

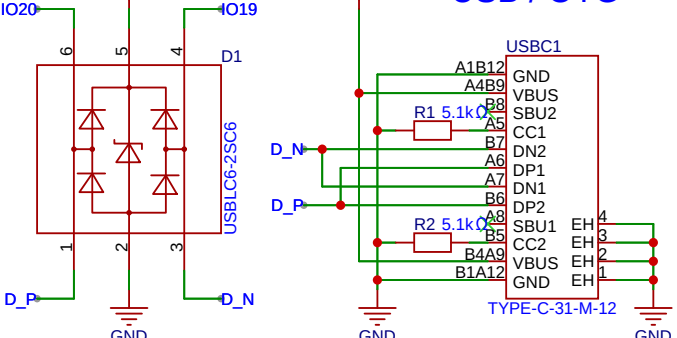


Camera: OV2640

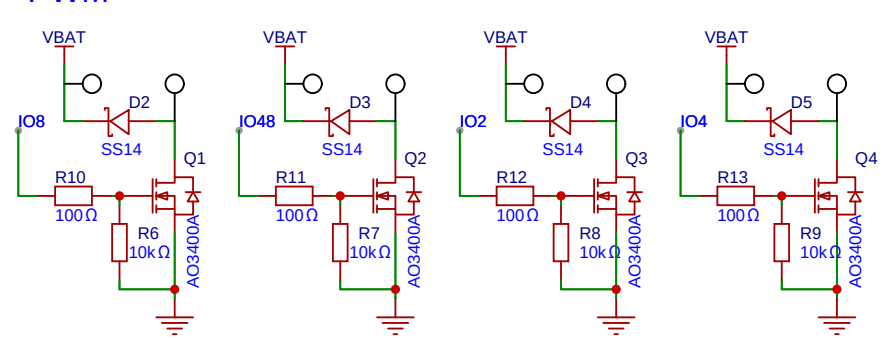
IMU



USB / OTG



PWM



Schematic	Schematic_TPJ01		Create at	2026-05-24
Board	TPJ-01		Update at	2026-06-15
Drawn			Page	P1
Reviewed			Tapejara_tpj01_v1	
		Version	Size	Page 1 Total 1
		V1.0	A3	UFABC



